```
# Importing the Necessary Libraries

import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
```

```
# 1.Load a dataset in your IDE

dataset = pd.read_csv('https://raw.githubusercontent.com/datasciencedojo/datasets/master/titanic.csv')
```

```
# 2.Observe the statistics of all the features

dataset.describe()
```

	PassengerId	Survived	Pclass	Age	SibSp	Parch	Fare
count	891.000000	891.000000	891.000000	714.000000	891.000000	891.000000	891.000000
mean	446.000000	0.383838	2.308642	29.699118	0.523008	0.381594	32.204208
std	257.353842	0.486592	0.836071	14.526497	1.102743	0.806057	49.693429
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000
25%	223.500000	0.000000	2.000000	20.125000	0.000000	0.000000	7.910400
50%	446.000000	0.000000	3.000000	28.000000	0.000000	0.000000	14.454200
75%	668.500000	1.000000	3.000000	38.000000	1.000000	0.000000	31.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.000000	512.329200

```
# 3.Obtain the shape of the dataset

dataset.shape
```

```
(891, 12)
```

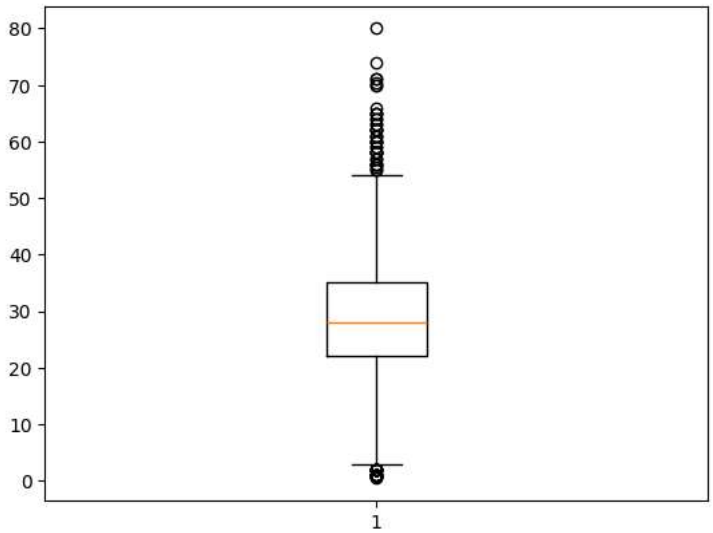
```
# 4.Separate all the features

dataset.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   PassengerId  891 non-null    int64
 1   Survived     891 non-null    int64
 2   Pclass       891 non-null    int64
 3   Name         891 non-null    object
 4   Sex          891 non-null    object
 5   Age          714 non-null    float64
 6   SibSp        891 non-null    int64
 7   Parch        891 non-null    int64
 8   Ticket       891 non-null    object
 9   Fare         891 non-null    float64
 10  Cabin        204 non-null    object
 11  Embarked     889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

```
# Drop rows where target 'Survived' is missing (none in this dataset, but just in case)
dataset = dataset.dropna(subset=["Survived"])

# Fill missing values properly (no inplace warning)
dataset["Age"] = dataset["Age"].fillna(dataset["Age"].median())
dataset["Embarked"] = dataset["Embarked"].fillna(dataset["Embarked"].mode()[0])
dataset["Fare"] = dataset["Fare"].fillna(dataset["Fare"].median())

# Drop Cabin (too many missing values)
dataset = dataset.drop(columns=["Cabin"])
```

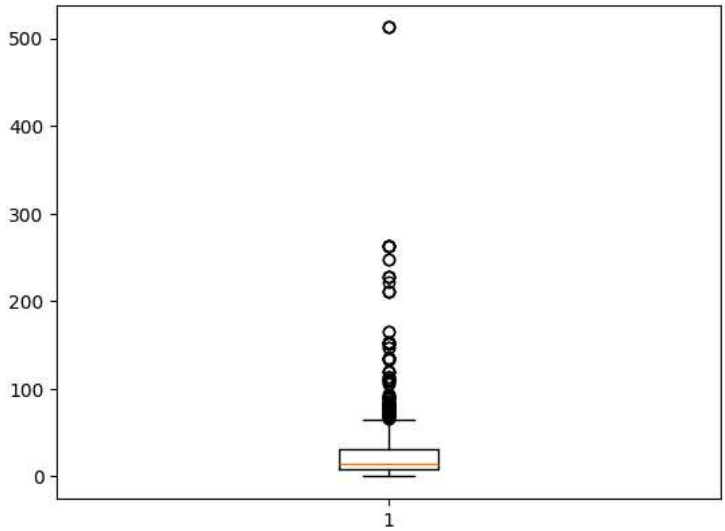
```
# 6.Observe the Box-Plot of each feature

# For Rating

plt.boxplot(dataset['Age'])
```

```
{'whiskers': [<matplotlib.lines.Line2D at 0x7aaee86a5c10>,
  <matplotlib.lines.Line2D at 0x7aaee86ad310>],
 'caps': [<matplotlib.lines.Line2D at 0x7aaee86ad640>,
  <matplotlib.lines.Line2D at 0x7aaee86ad9a0>],
 'boxes': [<matplotlib.lines.Line2D at 0x7aaee86a63c0>],
 'medians': [<matplotlib.lines.Line2D at 0x7aaee86adcd0>],
 'fliers': [<matplotlib.lines.Line2D at 0x7aaee86ae030>],
 'means': []}
```




```
# For Size

plt.boxplot(dataset['Fare'])
```

```
{'whiskers': [<matplotlib.lines.Line2D at 0x7aaeb5635b50>,
  <matplotlib.lines.Line2D at 0x7aaeb5635e50>],
 'caps': [<matplotlib.lines.Line2D at 0x7aaeb5636180>,
  <matplotlib.lines.Line2D at 0x7aaeb5636540>],
 'boxes': [<matplotlib.lines.Line2D at 0x7aaeb56357f0>],
 'medians': [<matplotlib.lines.Line2D at 0x7aaeb56368d0>],
 'fliers': [<matplotlib.lines.Line2D at 0x7aaeb5636c60>],
 'means': []}
```



Start coding or generate with AI.